

CONTACT INFORMATION	Department of Economics, Management & Statistics University of Milano-Bicocca Via Bicocca degli Arcimboldi, 8, 20126, Milan, Italy Email: tommaso.rigon@unimib.it Website: tommasorigon.github.io Orcid ID: orcid.org/0000-0002-9224-543X
RESEARCH INTERESTS	Applied Bayesian Modeling, Bayesian clustering, Bayesian Nonparametrics, Computational Statistics, Functional Data Analysis, Mixture Models, Species Sampling Models
RESEARCH NETWORK	Member of BayesLab at the Bocconi Institute for Data Science and Analytics (BIDSA), and the MIDAS Complex Data Modeling Research Network
CURRENT POSITION	Associate Professor (<i>Professore Associato, STAT-01/A</i>) University of Milano-Bicocca, Department of Economics, Management & Statistics (DEMS), Milan, Italy (from 05/2026 to present)
PAST ACADEMIC POSITIONS	Senior Assistant Professor (<i>Ricercatore, SECS-S/01, Legge 240/10 tipo B</i>) University of Milano-Bicocca, Department of Economics, Management & Statistics (DEMS), Milan, Italy (from 05/2023 to 04/2026) Junior Assistant Professor (<i>Ricercatore, SECS-S/01, Legge 240/10 tipo A</i>) University of Milano-Bicocca, Department of Economics, Management & Statistics (DEMS), Milan, Italy (from 10/2020 to 04/2023) Postdoctoral Associate Duke University, Department of Statistical Sciences, Durham, USA (from 02/2020 to 09/2020) Research Associate Duke University, Department of Statistical Sciences, Durham, USA (from 10/2019 to 02/2020) Research Affiliate de Castro Statistics Initiative, Collegio Carlo Alberto, Turin, Italy (from 09/2017 to 09/2020)
EDUCATION	Bocconi University, Milan, Italy Ph.D. in Statistics (<i>Dottorato di ricerca in Statistica</i>), Department of Decision Sciences (from 09/2015 to 09/2019, thesis defense 01/2020) – Thesis topic: <i>Finite-dimensional nonparametric priors: theory and applications</i> – Advisors: Antonio Lijoi and Igor Prünster – Ph.D. awarded with honors Università degli Studi di Padova, Padova, Italy M.Sc. in Statistical Sciences (<i>Laurea Magistrale in Scienze Statistiche</i>), School of Statistics (from 10/2013 to 09/2015) – Thesis topic: <i>Functional telecommunication data: a Bayesian nonparametric approach</i> – Advisor: Bruno Scarpa – Final mark: 110/110 with honors B.Sc. in Statistics, Economics & Finance (<i>Laurea Triennale in Statistica, Economia & Finanza</i>), School of Statistics (from 10/2010 to 04/2013) – Thesis topic: <i>Box-Cox transformation: an analysis based on the likelihood</i> – Advisor: Nicola Sartori – Final mark: 110/110 with honors
AWARDS	Academic awards 2026 <i>National Scientific Qualification (ASN) for Full Professor in Statistics (13/D1)</i>. Awarded by the Italian Ministry of University and Research 2024 <i>Blackwell–Rosenbluth Award 2024</i>. Junior Section of the International Society for Bayesian Analysis (J-ISBA) 2024 <i>Mitchell Prize 2023</i>. American Statistical Association (ASA) & International Society for Bayesian Analysis (ISBA) 2022 <i>National Scientific Qualification (ASN) for Associate Professor in Statistics (13/D1)</i>. Awarded by the Italian Ministry of University and Research 2021 <i>Young Talents Award</i>. University of Milano-Bicocca & Accademia Nazionale dei Lincei 2021 <i>Savage Award 2020 (Theory and Methods)</i>. American Statistical Association (ASA) & International Society for Bayesian Analysis (ISBA)

2017 *Best Ph.D. student in Statistics*. Bocconi University

Data competitions

2018 *Best objective prediction*. Stats under the Stars 4

2017 *Young-CLADAG data contest*. Classification and Data Analysis group

2017 *Bocconi summer school data competition*. Bocconi summer school

Travel awards

2019 *Travel award (accommodation)*. BNP12 conference, Oxford, UK

2019 *Travel award (400€)*. O'Bayes 2019 conference, Warwick, UK

2018 *ISBA travel award (250\$)*. ISBA 2018 world meeting, Edinburgh, UK

2017 *ISBA travel award (700\$)*. O'Bayes 2017 conference, Austin, Texas

SERVICE TO PROFESSION

Editorial activity

2022 Associate Editor of *Bayesian Analysis* (from 2022 to present)

Referee service (alphabetical order)

Annals of Applied Statistics · Annals of Statistics · Bayesian Analysis · Bernoulli · Biometrics · Biometrika · Computational Statistics & Data Analysis · Electronic Journal of Statistics · European Journal of Operational Research · IEEE Transactions on Neural Networks and Learning Systems · Journal of Computational & Graphical Statistics · Journal of the American Statistical Association · Journal of the Royal Statistical Society Series B · Journal of Statistical Planning and Inference · NeurIPS · PLOS ONE · Social Indicators Research · Statistical Methods & Applications · Statistical Science · Statistics & Probability Letters

Positions in academic societies

2026/27 **Program Chair** of the [Bayesian Nonparametric Section](#) of the International Society for Bayesian Analysis

2024 **Peer mentor** in the [j-ISBA Peer Mentoring Scheme](#), “Researchers supporting other researchers”

2024 **Member of the committee** for the DeGroot Prize

2022 **Member of the committee** for the Savage Award, Theory and Methods

2021/24 **Social media manager** of the International Society for Bayesian Analysis

2016/18 **Elected member of the board** of the young group (ysis) of the Italian Statistical Society

Organization of scientific events

2024 **Member of the local and scientific committees** of the *BAYesian Young Statisticians Meeting 2024* (BAYSM 2024), Venice, Italy

2022 **Organizer of the session** *Flexible Bayesian modelling for biostatistics* at the [15th International Conference of the ERCIM WG on Computational and Methodological Statistics](#), London, UK

2021 **Organizer of the j-ISBA session** *Computational and modeling advances for complex biological data* at the [2021 World Meeting of the International Society for Bayesian Analysis](#), virtual

2019 **Organizer of the ysis session** *Think outside of the black-box: statistical reasoning in applications* at the [sis2019: Smart Statistics for Smart Applications](#), Milan, Italy

2019 **Member of the organizing committee** of *Stats under the Stars 5*, a hackathon for young Data Scientists, Milan, Italy

2017 **Member of the local organizing committee** of the *10th international workshop on Bayesian inference in stochastic processes* (BISP10), Milan, Italy

SERVICE TO UNIVERSITY

Positions in boards and committees

– **Faculty member of the Ph.D. program** in *Economics, Statistics and Data Science*, University of Milano-Bicocca (from 2024 to present)

– **Member of the admission committee for the Ph.D. program** in *Economics, Statistics and Data Science*, University of Milano-Bicocca (2024)

– **Member of the selection committee for adjunct professors** at the University of Milano-Bicocca (5+ times, 2023–2025)

– **Doctoral thesis committee member** for

· *Ph.D. in Mathematical Models and Methods in Engineering* at Politecnico di Milano (Alessandro Carminati, Michela Frigeri, 2026)

· *Ph.D. in Statistics and Computer Science* at Bocconi University (Patric Dolmeta, 2024)

· *Ph.D. in Economics and Statistics* at the University of Milano-Bicocca (Camilla Salvatore, Luca Brusa, 2023; Luca Aiello, 2024)

· *Ph.D. in Mathematics, Information sciences and technologies, Computer science* at the University Grenoble Alpes (Louise Alamichel, 2024)

- **Social media manager of the Department** (DEMS), University of Milano-Bicocca, from 2021 to present

Students supervised, co-supervised, and mentored

- **Current Ph.D. students:** Anna Petranzan (Ph.D.)
- **Former Ph.D. students:** Luca Presicce (Ph.D.), Alessandro Zito (Ph.D., supervisor: David Dunson, 2024), Davide Agnoletto (M.Sc., Ph.D., supervisor: Bruno Scarpa), Riccardo Cogo (Ph.D., supervisor: Federico Camerlenghi), Ching-Lung Hsu (Ph.D., supervisor: David Dunson)
- About 30 B.Sc./M.Sc. students supervised and co-supervised in applied and research-oriented theses on a broad range of topics

FUNDING AND GRANTS

Collaborator of the NEMESIS ERC project entitled “*sociogeNEsis of criMinal nEtworks: reconStruction, dIScovery and diSruption*”, funded by the European Research Council ERC starting grant for the years 2024–2029. PI: Daniele Durante

Member of the research group of the grant PRIN 2022 entitled “*Measuring Biodiversity via Bayesian Nonparametrics: Estimation, Clustering and Uncertainty Quantification*” for the years 2023–2025. PI: Igor Prünster. CO-PI: Federico Camerlenghi

Member of the research group of the grant PRIN 2022 entitled “*Discrete random structures for Bayesian learning and prediction*” for the years 2023–2025. PI: Antonio Lijoi. CO-PI: Bernardo Nipoti

Member of the research group of the grant FAQC 2021 funded by the University of Milano-Bicocca. PI: Bernardo Nipoti

Member of the research group of the LIFEPLAN ERC project, funded by the European Research Council through a 12.6M€ ERC-synergy grant for the years 2020–2026. Global biodiversity research PI: Tomas Roslin. Statistical research PI: David B. Dunson. Corresponding PI: Otso Ovaskainen

Member of the research group of the grant R01ES027498 funded by the National Institute of Environmental Health Sciences of the United States National Institutes of Health. PI: Amy H. Herring

PUBLICATIONS

Refereed journals

1. Anceschi, N., Castiglione, C., **Rigon, T.**, Zanella, G. and Durante, D. (2026+). Optimal and computationally tractable lower bounds for logistic log-likelihoods. *Biometrika*, to appear
2. Agnoletto, D., **Rigon, T.** and Dunson, D. B. (2026+). Nonparametric predictive inference for discrete data via Metropolis-adjusted Dirichlet sequences. *Journal of the American Statistical Association (T&M)*, to appear
3. Ghilotti, L., Camerlenghi, F. and **Rigon, T.** (2026). Bayesian analysis of product feature allocation models. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, **88**(2), 540–566
4. Zito, A., **Rigon, T.** and Dunson, D. B. (2026). Bayesian nonparametric modeling of latent partitions via Stirling-gamma priors. *Bayesian Analysis*, **21**(1), 139–166
5. Agnoletto, D., **Rigon, T.** and Dunson, D. B. (2025). Bayesian inference for generalized linear models via quasi-posteriors. *Biometrika*, **112**(2)
6. **Rigon, T.**, Scarpa, B. and Petrone, S. (2025). Enriched Pitman–Yor processes. *Scandinavian Journal of Statistics*, **52**(2), 631–657
7. Lijoi, A., Prünster, I. and **Rigon, T.** (2024). Finite-dimensional discrete random structures and Bayesian clustering. *Journal of the American Statistical Association (T&M)*, **119**(546), 929–941
8. Catalano, M., Lijoi, A., Prünster, I. and **Rigon, T.** (2023). Bayesian modeling via discrete nonparametric priors. *Japanese Journal of Statistics and Data Science*, **6**(2), 607–624
9. **Rigon, T.** (2023). An enriched mixture model for functional clustering. *Applied Stochastic Models in Business and Industry*, **39**(2), 232–250
10. **Rigon, T.** and Aliverti, E. (2023). Conjugate priors and bias reduction for logistic regression models. *Statistics & Probability Letters*, **202**, 109901
11. **Rigon, T.**, Herring, A. H. and Dunson, D. B. (2023). A generalized Bayes framework for probabilistic clustering. *Biometrika*, **110**(3), 559–578
12. Zito, A., **Rigon, T.** and Dunson, D. B. (2023). Inferring taxonomic placement from DNA barcoding aiding in discovery of new taxa. *Methods in Ecology & Evolution*, **14**(2), 529–542
13. Zito, A., **Rigon, T.**, Ovaskainen, O. and Dunson, D. B. (2023). Bayesian modelling of sequential discoveries. *Journal of the American Statistical Association (T&M)*, **118**(544), 2521–2532
14. Legramanti, S., **Rigon, T.** and Durante, D. (2022). Bayesian testing for exogenous partition structures in stochastic block models. *Sankhyā A: The Indian Journal of Statistics*, **84**, 108–126

15. Legramanti, S., **Rigon, T.**, Durante, D. and Dunson, D. B. (2022). Extended stochastic block models with application to criminal networks. *Annals of Applied Statistics*, **16**(4), 2369–2395
16. Reverberi, C., **Rigon, T.**, Solari, A., Hassan, C., Cherubini, P., GI Genius CADx Study Group and Cherubini, A. (2022). Experimental evidence of effective human-AI collaboration in medical decision-making. *Scientific Reports*, **12**, 14952
17. Favaro, S., Panero, F. and **Rigon, T.** (2021). Bayesian nonparametric disclosure risk assessment. *Electronic Journal of Statistics*, **15**(2), 5626–5651
18. **Rigon, T.** and Durante, D. (2021). Tractable Bayesian density regression via logit stick-breaking priors. *Journal of Statistical Planning and Inference*, **211**, 131–142
19. Lijoi, A., Prünster, I. and **Rigon, T.** (2020). Sampling hierarchies of discrete random structures. *Statistics & Computing*, **30**, 1591–1607
20. Lijoi, A., Prünster, I. and **Rigon, T.** (2020). The Pitman–Yor multinomial process for mixture modeling. *Biometrika*, **107**(4), 891–906
21. Durante, D. and **Rigon, T.** (2019). Conditionally conjugate mean-field variational Bayes for logistic models. *Statistical Science*, **34**(3), 472–485
22. **Rigon, T.**, Durante, D. and Torelli, N. (2019). Bayesian semiparametric modelling of contraceptive behavior in India via sequential logistic regressions. *Journal of the Royal Statistical Society Series A: Statistics in Society*, **182**(1), 225–247
23. Durante, D., Canale, A. and **Rigon, T.** (2019). A nested expectation-maximization algorithm for latent class models with covariates. *Statistics & Probability Letters*, **146**, 97–103

Submitted and working papers

24. Stolf, F., **Rigon, T.**, and Dunson, D. B. (2026+). Bayesian inference on beta diversity via feature allocation models with imperfect detection. Submitted. *arXiv:2608.11180*
25. Zito, A., **Rigon, T.**, Roslin, T., Niittynen, P., Hebert, P. D. N., Zakharov, E., Ratnasingham, S., iBOL Consortium, Ovaskainen, O. and Dunson, D. B. (2026+). Predicting global biodiversity via Hubbell regression. Submitted. *bioRxiv:2026.05.29.728852*
26. Ghilotti, L., Camerlenghi, F., **Rigon, T.** and Guindani, M. (2026+). Bayesian nonparametric modeling of multivariate count data with an unknown number of traits. Submitted. *arXiv:2510.24526*
27. **Rigon, T.**, Hsu, C. and Dunson, D. B. (2026+). A Bayesian theory for estimation of biodiversity. Submitted. *arXiv:2502.01333*

Publications in monographs, volumes, and discussions

28. Agnoletto, D., **Rigon, T.** and Dunson, D. B. (2025). Bayesian inference for generalized linear models via quasi-posteriors: an application to Eurasian Chaffinch abundance in Finland. In *Methodological and Applied Statistics and Demography III* (Pollice, A. and Mariani, P., editors). Springer
29. **Rigon, T.**, Aliverti, E., Russo, M. and Scarpa, B. (2021). A discussion on: “Centered partition processes: Informative priors for clustering” by Paganin, S., Herring, A. H., Olshan, A. F., Dunson, D. B., et al. (2021) in *Bayesian Analysis* **16**(1), 301–370
30. Aliverti, E., Paganin, S., **Rigon, T.** and Russo, M. (2019). A discussion on: “Latent nested nonparametric priors” by Camerlenghi, F., Dunson, D. B., Lijoi, A., Prünster, I. and Rodríguez, A. in *Bayesian Analysis* **14**(4), 1303–1356
31. Caponera, A., Denti, F., **Rigon, T.**, Sottosanti, A. and Gelfand, A. (2018). Hierarchical spatio-temporal modeling of resting state fMRI data. In *Studies in Neural Data Science* (Canale, A., Durante, D., Paci, L. and Scarpa, B., editors). Springer

National conference proceedings

32. Agnoletto, D., **Rigon, T.** and Scarpa, B. (2023). Bayesian density estimation for modeling age-at-death distribution. In *Book of Short Papers of the Italian Statistical Society* (Chelli, F. M., Ciommi, M., Ingrassia, S., Mariani, F., Recchioni, M. C.) 2023. ISBN: 9788891935618
33. Cogo, R., Camerlenghi, F. and **Rigon, T.** (2023). Hierarchical processes in survival analysis. In *Book of Short Papers of the Italian Statistical Society* (Chelli, F. M., Ciommi, M., Ingrassia, S., Mariani, F., Recchioni, M. C.) 2023. ISBN: 9788891935618
34. Presicce, L., **Rigon, T.** and Aliverti, E. (2023). Bias-reduction methods for Poisson regression models. In *Book of Short Papers of the Italian Statistical Society* (Chelli, F. M., Ciommi, M., Ingrassia, S., Mariani, F., Recchioni, M. C.) 2023. ISBN: 9788891935618
35. Legramanti, S., **Rigon, T.** and Durante, D. (2022). Bayesian clustering of brain regions via extended stochastic block models. In *Book of Short Papers of the Italian Statistical Society* (Balzanella, A., Bini, M., Cavicchia, C.,

Verde, R.) 2022. ISBN: 9788891932310

36. Zito, A., **Rigon, T.** and Dunson, D. B. (2021). Modelling of accumulation curves through Weibull survival functions. In *Book of Short Papers of the Italian Statistical Society 2021* (Perna, C., Salvati, N. and Schirripa Spagnolo, F.). ISBN: 9788891927361
37. **Rigon, T.** (2018). Logit stick-breaking priors for partially exchangeable count data. In *Book of Short Papers of the Italian Statistical Society 2018* (Abbruzzo, A., Piacentino, D., Chiodi, M. and Brentari, E.). ISBN: 9788891910233

VISITING PERIODS **Universities and research centers**

- 2023 *Duke University*, North Carolina, USA (April 12th, 2023 – May 9th, 2023)
2022 *Inria centre at the University Grenoble Alpes*, France (April 3rd, 2022 – April 10th, 2022)

PRESENTATIONS **Seminars**

- 2026 *Università di Roma, La Sapienza*, Rome, Italy
2024 *Inria centre at the University Grenoble Alpes*, Grenoble, France
2023 *University of California in Los Angeles (UCLA)*, California, USA
2023 *Duke University*, North Carolina, USA
2023 *Università Ca' Foscari*, Venezia, Italy
2021 *University of Crete*, Greece
2020 *Collegio Carlo Alberto, Università degli Studi di Torino*, Italy

Keynote and plenary presentations

- 2025 *sis Bayes 2025*, University of Padova, Padova, Italy

Invited presentations

- 2026 *2026 ISBA world meeting*, Nagoya, Japan
2026 *Workshop on Advances in BAYesian COmputation and modeling (ABACO26)*, University of Padova, Padova, Italy
2025 *GRASPA environmental statistics 2025*, University of Roma Tre and LUMSA University, Rome, Italy
2025 *14th International Conference on Bayesian Nonparametrics*, UCLA, Los Angeles, USA
2025 *Early Career Workshop on Nonparametric Statistics*, LUISS, Rome, Italy
2025 *14th Bayesian Inference for Stochastic Processes*, Milan, Italy
2025 *BAYSM 2025: Bayesian Young Statisticians Meeting (Blackwell–Rosenbluth Award Talks)*, online
2024 *BIRS-CMO 2024: Frontiers of Bayesian Inference and Data Science*, Oaxaca, Mexico (hybrid)
2024 *Bernoulli-IMS 11th World Congress in Probability and Statistics*, Bochum, Germany
2024 *2024 ISBA world meeting (Biometrika invited session)*, Venice, Italy
2024 *ISBA satellite workshop in Lugano*, Lugano, Switzerland
2023 *13th Bayesian Inference for Stochastic Processes*, Madrid, Spain
2023 *BayesComp 2023: Bayesian computing without exact likelihoods (satellite event)*, Levi, Finland
2022 *The IISA International Conference on Statistics*, Bangalore, India
2022 *15th International Conference of the ERCIM WG on Computational and Methodological Statistics*, London, UK
2022 *Statistical methods and models for complex data (invited discussant)*, Padova, Italy
2022 *CSDA & EcoSta Workshop on Statistical Data Science (SDS 2022)*, Bologna, Italy
2022 *5th International Conference on Econometrics and Statistics*, Tokyo, Japan (hybrid)
2022 *BNP Networking Event*, Nicosia, Cyprus
2021 *14th International Conference of the ERCIM WG on Computational and Methodological Statistics*, London, UK (hybrid)
2021 *2021 ISBA world meeting*, virtual
2020 *Bayes Comp 2020*, Gainesville, Florida, USA
2019 *The IISA International Conference on Statistics*, Mumbai, India
2019 *12th International Conference of the ERCIM WG on Computational and Methodological Statistics*, London, UK
2018 *11th International Conference of the ERCIM WG on Computational and Methodological Statistics*, Pisa, Italy
2018 *JSM 2018: Joint Statistical Meetings*, Vancouver, Canada
2018 *49th Scientific meeting of the Italian Statistical Society*, Palermo, Italy
2017 *The IISA International Conference on Statistics*, Hyderabad, India

Contributed presentations

- 2026 *BNP Networking Event*, Seoul, South Korea

2024 *BNP Networking Event*, Singapore
 2023 *Approximation methods in Bayesian Analysis*, CIRM, Marseille, France
 2019 *12th International Conference on Bayesian Nonparametrics*, Oxford, UK
 2018 *BAYSM 2018: Bayesian Young Statisticians Meeting*, Warwick, UK
 2018 *YES IX Scalable Statistics: on Accuracy and Computational Complexity*, Eindhoven, Netherlands

Poster presentations

2019 *O'Bayes 2019*, Warwick, UK
 2018 *2018 ISBA world meeting*, Edinburgh, UK
 2017 *O'Bayes 2017*, Austin, USA
 2017 *10th international workshop on Bayesian inference in stochastic processes (BISP10)*, Milan, Italy
 2017 *Italian Statistical Society (sis) Bayes 2017*, Rome, Italy
 2016 *2016 ISBA world meeting*, Cagliari, Italy

TEACHING

University of Milano-Bicocca, Milan, Italy *As lecturer*

- Statistical Inference (Ph.D.) – A.Y. 2024/25, 2025/26, 32h
- Bayesian Computations (Ph.D.) – A.Y. 2020/21, 2021/22, 12h; A.Y. 2022/23, 2023/24, 2024/25, 15h
- Data Mining (M.Sc.) – A.Y. 2023/24, 42h; A.Y. 2024/25, 2025/26, 47h
- Statistics III (B.Sc.) – A.Y. 2025/26, 40h
- Statistics I (B.Sc.) – A.Y. 2020/21, 2021/22, 2022/23, 42h; A.Y. 2023/24, 2024/25, 48h
- R for the multivariate statistical analysis (B.Sc.) – A.Y. 2020/21, 2021/22, 2022/23, 21h

SDA Bocconi School of Management, Milan, Italy *As lecturer*

- AXA Italia – Mastering data for insurance, III ed. – A.Y. 2022/23, 12h
- AXA Italia – Mastering data for insurance, II ed. – A.Y. 2021/22, 8h

Bocconi University, Milan, Italy *As lecturer*

- Foundations of data science (B.Sc.) – A.Y. 2022/23, 2023/24, 2024/25, 10h

As teaching assistant

- Data analysis (B.Sc.) – A.Y. 2016/17
- Statistics for economics and business (M.Sc.) – A.Y. 2016/17

Università degli Studi di Padova, Padova, Italy *As lecturer*

- Data mining (M.Sc.) – A.Y. 2020/21, 8h

As teaching assistant

- Introduction to real analysis (B.Sc.) – A.Y. 2014/15
- Advanced statistical inference (M.Sc.) – A.Y. 2014/15

PAST BUSINESS POSITIONS

Intern

Groupon, Dublin, Ireland (from 09/2014 to 11/2014)

Business analyst

Bravofly Rumbo Group, Chiasso, Switzerland (from 10/2013 to 07/2014)

Intern

Bravofly Rumbo Group, Chiasso, Switzerland (from 05/2013 to 10/2013)

Le dichiarazioni rese nel presente curriculum sono da ritenersi rilasciate ai sensi degli artt. 46 e 47 del D.P.R. 445/2000.

Autorizzo la pubblicazione ai sensi del D.Lgs. n. 33/2013 e il trattamento dei dati personali ai sensi del Regolamento (UE) 2016/679 e del D.Lgs. n. 196/2003.